GEFFEN COOPER

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PhD candidate in Electrical & Computer Engineering at UT Austin with industry experience in embedded systems and edge ML. During my PhD, I built a solar-powered wearable sensing system, integrating a custom PCB, energy-aware firmware for batteryless operation and BLE communication, and a multisensor data collection pipeline. I have also deployed production ML models at Wiliot and tools to monitor performance across 1,000+ stores.

EDUCATION

The University of Texas at Austin Aug. 2022 - present
Ph.D. in Electrical and Computer Engineering (GPA: 3.9/4.0) *Expected: March 2027*

University of California, Santa Barbara Sept. 2018 - June 2022
B.S. in Computer Engineering (GPA: 3.97/4.0)

RELEVANT WORK EXPERIENCE

UT Austin August 2022 - Present
Graduate Research Assistant, System Level Design Group *Austin, TX*

- ***Batteryless* sensing system:** designed and built a solar-powered wearable accelerometer, including custom PCB, firmware, and data collection infrastructure for multisensor human activity recognition.
- Developed data acquisition and processing algorithms for sparse, asynchronous accelerometer time-series, enabling sensing and inference under dynamic energy harvesting conditions.
- Conducted an IRB-approved study to collect and release a batteryless human activity recognition dataset .

Wiliot June 2025 - May 2026
Data Scientist *Remote*

- Redesigned an XGBoost classifier using RSSI and location data from batteryless BLE pixels, reducing the feature count by 5× while improving interpretability, training time (2×), and accuracy on new stores ($> 5\%$).
- Developed a retrospective monitoring system to track the performance of a production ML model across 1,000+ stores without ground-truth labels.

Analog Devices June 2022 - August 2022
Embedded AI Applications Intern, Embedded AI Team *Dallas, TX*

- Implemented a few-shot domain adaptation technique to fine-tune CNN models for the MAX78000 MCU
- Adapted pretrained models using 5-10 images per class from the onboard camera to improve on-device classification accuracy on camera images by 20-40%

Brain Corp June 2020 - September 2020
Embedded Software Intern, Firmware Team *San Diego, CA*

- Created a robust Linux based C/C++ application using socketCAN to update the firmware of the vehicle controller used in Brain Corp's fleet of autonomous floor scrubbing robots

FLIR Systems April 2019 - August 2019
Software Intern, Security Team *Goleta, CA*

- Wrote bash scripts and python GUI for running diagnostics, debugging, and configuring thermal cameras

TECHNICAL SKILLS

Languages & Tools: Python, C, Bash, Git, SQL, PySpark
Embedded & IoT: nRF52 Series, Zephyr RTOS, BLE, SPI, SWD, KiCad, Logic analyzer, Oscilloscope
Machine Learning: PyTorch, TensorFlow Lite, ARM CMSIS-NN

SELECTED PUBLICATIONS

Geffen Cooper, N. Tsao, F. Fotiadis, U. Topcu, and R. Marculescu, "BHAR: A System-level Optimization Framework for Multisensor **B**atteryless **H**uman **A**ctivity **R**ecognition," ***Accepted*** at 2026 IEEE/ACM International Conference on Codesign of Embedded Systems (CODES).

Geffen Cooper, Radu Marculescu, "Batteryless Gesture Recognition via Learned Sampling," IEEE International Conference on Body Sensor Networks (BSN 2025).

Geffen Cooper, Radu Marculescu, "Packet Pruning: Finding Better Energy Spending Policies for Batteryless Human Activity Recognition," IEEE International Conference on Body Sensor Networks (BSN 2024).

Geffen Cooper, Radu Marculescu, "Beyond Thresholds: A General Approach to Sensor Selection for Practical Deep Learning-based HAR," IEEE/ACM International Conference on Internet-of-Things Design and Implementation (IoTDI 2024).

Geffen Cooper, Tianda Huang, Radu Marculescu, "Demo Abstract: A Prototype for Machine Learning with Batteryless Sensors," IEEE/ACM International Conference on Internet-of-Things Design and Implementation (IoTDI 2024).

ADDITIONAL PROJECTS

TinyML Projects with Max78000 Collaboration with Analog Devices 2021-2022

- Deployed image classification and detection models with MAX78000 AI microcontroller with CNN accelerator
- **G. Cooper**, V. Benenati, B. Long, K. Copeland, T. Ekaireb, S. Kumar, B.S. Manjunath, Y. Isukapalli, "Autonomous System for Sorting Objects at The Edge," Proceedings, International Telemetering Conference (ITC), Glendale, Arizona, Oct 24-27, 2022.
- **G. Cooper**, B.S. Manjunath, and Y. Isukapalli, "Edge Machine Learning for Face Detection," Proceedings, International Telemetering Conference (ITC), Las Vegas, Nevada, Oct 25-28, 2021.